12TH CHEMISTRY · CHAPTER 31 · FULL CHAPTER NOTES

Materials and Energy

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] Why were new materials and energy devices needed?

Humans have long used natural resources such as wood, metals and materials obtained from animals. With the enormous increase in population and limited natural resources, new materials and machines were needed. This led to the invention of many materials, from metal alloys to nanomaterials and from large electric cells to very efficient miniature batteries. Modern machines (computers, mobiles, construction machinery, aircraft) all combine modern materials and energy devices; e.g. a mobile phone contains metals, plastics, composites and a small but efficient Li-ion battery.

31.1 MATERIALS

Q.2 [Short] What is a material? On what basis is a material selected for an application?

A material is any substance or mixture of substances that makes up a physical object of daily-life or industrial importance, e.g. plastics, metals, ceramics and building materials. Materials are understood by their composition, the nature of their chemical bonds and their microstructure. A material is selected for an application according to its physical, chemical, mechanical, electrical and thermal characteristics.

Q.3 [Short] What are the properties of metals? How are they applied?

Metals are hard, rigid substances that are good conductors of heat and electricity. They are malleable and ductile, so they can be rolled into sheets or drawn into wires without fracture. Steel is used in bridges and buildings because of its strength; aluminium in aircraft bodies because it is light-weight; copper in electric wires because of its excellent conductivity.

Q.4 [Short] How do metals occur in nature? What is gangue?

Metals are mostly found in the earth’s crust. Less reactive metals such as gold and platinum occur in the free (metallic) state. More reactive metals such as sodium, aluminium, iron and copper exist in the combined state as ores, e.g. haematite (Fe_2O_3) is an ore of iron. Ores are not pure compounds; they are mixtures of the desired metal compound with impurities and earthy materials called **gangue**.

Q.5 [Long] What is metallurgy? Explain the steps in the extraction of metals from their ores.

Metallurgy is the science of extracting metals from their ores and making them pure for better use. The reactivity and nature of the metal decide the extraction procedure.

i. Mining: acquiring the metal ores that are buried deep in the earth, e.g. iron ores.

ii. Ore concentration: the ore is made free from impurities (gangue) as much as possible.

iii. Conversion of the concentrated ore to metal oxide: by *roasting* (heating a sulphide ore with an excess supply of air) or by *calcination* (heating a carbonate ore in limited or no air).

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)} \text{ (roasting)}$$
$$ZnCO_{3(s)} \rightarrow ZnO_{(s)} + CO_{2(g)} \text{ (calcination)}$$

iv. Reduction of the metal oxide to metal: for less reactive metals (below carbon in the reactivity series) carbon is the reducing agent. Carbon is first converted into CO, which displaces oxygen from the metal oxide.

$$C + \tfrac{1}{2}O_2 \rightarrow CO$$
$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Metals more reactive than carbon (Na, K, Al) are obtained by the *electrolytic method*: the metal compound is melted and an electric current is passed; the positive metal ions move to the cathode and gain electrons to form the metal. It needs much energy and is also used to get very high purity in less reactive metals (e.g. Cu).

v. Refining: the final purification of the metal by thermal, chemical or electrochemical methods; e.g. copper must be 99.99% pure for electrical wiring.

vi. Alloying: mixing the metal with other metals or a non-metal to improve its properties.

Q.6 [Short] Differentiate between roasting and calcination.

Roasting	Calcination
Used for sulphide ores.	Used for carbonate ores.
Ore is heated with an excess supply of air.	Ore is heated in limited or no air.
SO_2 gas is given off: $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$	CO_2 gas is given off: $ZnCO_3 \rightarrow ZnO + CO_2$

Q.7 [Short] Why can carbon not be used to extract sodium or aluminium?

Carbon can reduce only the oxides of metals that are less reactive than itself (below carbon in the reactivity series). Na, K and Al are more reactive than carbon, so carbon cannot remove oxygen from their oxides. They are extracted by electrolysis of their molten compounds.

Q.8 [Long] What is alloying? Give the composition, properties and uses of common alloys.

Alloying is the mixing of two or more metals to give a homogeneous solution, or the mixing of a metal with a non-metal. It alters the microstructure of the metal to enhance properties such as strength, corrosion resistance and electrical or magnetic properties.

Alloy	Constituent elements	Properties	Common uses
Brass	Copper (60–90%) + zinc (10–40%)	Strong, corrosion-resistant	Electrical fittings
Bronze	Copper (88%) + tin (12%)	Hard, wear-resistant	Medals, statues
Steel	Iron (98–99%) + carbon (0.1–2%) (and Cr, Ni, Mn, Si, S)	Tough, durable, strong	Buildings, tools
Duralumin	Aluminium (93–95%) + copper (3.5–4.5%) + magnesium (0.4–0.7%) + manganese (0.4–0.7%)	Lightweight, strong, corrosion-resistant	Aircraft body

Q.9 [Long] What are ceramics? Give their composition, properties and applications.

Ceramics are hard but brittle materials with high melting temperatures and very good insulating capabilities, which suit them to high-temperature and electrical applications. Chemically they are a blend of silicates and oxides of alkali and alkaline earth metals and aluminium (e.g. K_2O, Na_2O, CaO, aluminium silicate). Strong bonds and fixed structures give them strength, durability and resistance to corrosion and heat. Uses: pottery, furnaces, tiles and sanitary ware (porcelain), windows and optical fibres (glass), spark plugs (alumina).

Property	Description	Application example
Hardness and wear resistance	Extremely difficult to scratch or abrade.	Abrasives (silicon carbide in sandpaper) and cutting tools.
Electrical insulator	Do not conduct electricity (most).	Porcelain insulators on power lines.
Chemical inertness	Highly resistant to corrosion and chemical attack.	Ceramic coatings on chemical processing equipment.

Q.10 [Short] What are polymers as materials?

Polymers are large inorganic or organic molecules having unique properties, prepared from a variety of small molecules. Organic polymers are much more common, but inorganic polymers are more significant in certain applications. (Details are in Chapter 26.)

Q.11 [Long] What are composites? Explain the roles of matrix and reinforcement, with uses.

A composite is a combination of two or more materials that forms a product with better properties. One component is the **matrix**; the others, caged in the matrix, are the **reinforcement agents** (a composite may have more than one). Example: fibreglass has a polymer matrix reinforced with fibres of fabric, glass or carbon-based materials.

Matrix	Reinforcement
The component in which the other components are caged.	The component(s) caged inside the matrix.
Holds the reinforcement together.	Provides high strength to the material.
In fibreglass: a polymer.	In fibreglass: fibres of fabric, glass or carbon-based materials.

Composites have a good strength-to-weight ratio and strong resistance to corrosion, so they suit demanding applications. Boat hulls and car bodies are made of fibreglass (also used to insulate walls and ceilings from heat and cold); carbon fibre is widely used in aircraft and sports gear because of its lightness and durability.

Q.12 [Long] What are semiconductors? Give examples and applications.

Semiconductors conduct electricity only under special conditions and are sensitive to light, heat and voltage. Unlike in a conductor, the current through a semiconductor can be controlled as required, which makes them important in modern electronics.

- **Silicon:** the most common; used in microchips and solar panels because it is highly effective and abundant. A mobile SIM card is a silicon chip with copper connectors; silicon can conduct, store and process data.
- **Gallium arsenide (GaAs):** used in wireless communication, e.g. 5G internet devices.
- **Gallium nitride (GaN):** used in electric cars for fast charging due to its advanced conductivity.

Quick Check 31.1

Q.13 [Short] i. Why ceramics are used as insulators on power lines.

Because most ceramics (e.g. porcelain) do not conduct electricity; they are also hard, durable and resistant to heat and corrosion, so they last in the open air.

Q.14 [Short] ii. Why is silicon preferred for use in microchips and solar panels?

Silicon is a semiconductor that is highly effective and abundant; the current through it can be controlled, and it can conduct, store and process data.

Q.15 [Short] iii. What do reinforcement agents provide in composite materials?

They provide high strength to the material, while the matrix holds them together.

31.2 TOXICITY AND RECYCLING

Q.16 [Short] Define toxicity.

Toxicity is the extent to which a substance can harm an organism or the environment. A material may be toxic during its production, use and final disposal. It is measured by the extent and type of hazard the material may cause.

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Q.17 [Long] Explain the types of hazards posed by toxic materials.

Hazard	Meaning	Example
Health hazard	The substance damages health upon exposure.	Chromium is a carcinogen and mutagen and causes respiratory issues; some plastic additives cause hormonal problems.
Physical hazard	Harm due to physical or chemical properties, to buildings, humans, ecosystems and the environment.	Flammable or explosive materials can harm humans and forests or damage a building.
Environmental hazard	Long-term harm to aquatic life or the environment.	Lead, mercury and cadmium leach into soil and groundwater and harm plants, animals and fish.

Q.18 [Short] Why are heavy metals and plastics major toxic materials?

Heavy metals: lead (Pb), mercury (Hg) and cadmium (Cd) are found in electronics, batteries and fluorescent lights. They can get into ground water and from there transfer to living organisms.

Plastics: many plastics have toxic additives such as phthalates or bisphenol A (BPA), which interfere with hormones. Microplastics are now found in oceans, soil and even food, with long-term effects on human life and ecosystems; in humans they are found to be carcinogenic.

Q.19 [Short] What is recycling? Give its advantages.

Recycling is an important part of waste management, conservation of resources and energy saving. It minimises solid waste in landfills and preserves limited natural resources such as metals, oil and timber. It also saves energy: recycling aluminium consumes 95 percent less energy than recovering aluminium from the raw ore.

Q.20 [Long] Explain the challenges associated with recycling.

1. **Material separation:** most present-day products (mobile phones, laptops) are assembled from mixed materials that are not easily separated, such as plastic coatings on metals or multi-layered packaging films.
2. **Expensive reprocessing:** reprocessing involves sorting, cleaning and other methods and needs a lot of energy, so it is usually expensive and uneconomical, especially when the materials are available elsewhere at a lower price.
3. **Contamination:** recycled metals and plastics can be contaminated by dyes, chemicals or biological materials, which lowers their quality and safety. Contaminated plastic cannot be recycled for food-grade use.

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Q.21 [Long] Describe the methods and principles of recycling important materials.

Material	Recycling method	Chemical principle / focus
Plastics	Mechanical recycling (reprocessing)	Sorting, cleaning, melting and reforming. Limited by contamination and degradation of polymer chains.
Plastics	Chemical recycling (depolymerization)	Polymer chains are broken into the original monomers or simpler chemicals using heat, solvents or catalysts. Gives purer reactants.
Metals (steel, aluminium)	Thermal and chemical refining (metallurgy)	Melting and refining to remove impurities. Needs much energy, but far less than primary production.
Paper	Pulping and de-inking	Chemical and mechanical separation of fibres; pulp is treated with NaOH or Na_2CO_3 to remove inks and dyes.
Glass	Cullet preparation and melting	Cleaning, crushing and melting scrap glass (cullet) to form new glass. Chemically simpler as glass is a non-crystalline solid.

Quick Check 31.2

Q.22 [Short] a) Name two types of hazards caused by toxic materials.

Health hazards and environmental hazards (physical hazards are the third type).

Q.23 [Short] b) Which methods will you use for the recycling i) empty metal food cans ii) empty beverage glass bottles iii) shopping bags and toys

i) Thermal and chemical refining (metallurgy): melting and refining to remove impurities. **ii)** Cullet preparation and melting: cleaning, crushing and melting the scrap glass to form new glass. **iii)** Plastics: mechanical recycling (sorting, cleaning, melting and reforming) or chemical recycling (depolymerization into monomers).

Q.24 [Short] c) How do heavy metals enter the food chain?

Heavy metals such as lead, mercury and cadmium from discarded electronics, batteries and fluorescent lights leach into soil and ground water. From there they are taken up by plants and aquatic organisms such as fish, and pass on to the animals and humans that eat them.

Q.25 [Short] d) How does recycling help conserve natural resources?

Recycled materials are used again in place of fresh raw materials, so limited natural resources such as metals (ores), oil and timber are preserved. It also saves energy, e.g. recycling aluminium uses 95% less energy than extraction from the ore.

31.3 X-RAY CRYSTALLOGRAPHY

Q.26 [Short] What is X-ray crystallography? Why are X-rays suitable?

X-ray crystallography is a very strong analytical tool used to identify the three-dimensional positions of atoms in a crystalline solid. X-rays have a very small wavelength, so they can find the positions of electrons and atoms in a molecule or an ionic compound.

Q.27 [Long] How does X-ray diffraction give the structure of a crystal?

1. Solid materials consist of layers or strands of atoms. X-rays from the source fall on the crystal.
2. The rays are diffracted after hitting the atoms at specific angles (path length difference = $2d \sin\theta$).
3. The diffracted rays form patterns on a screen, called diffraction patterns.
4. A diffraction pattern spots the location of atoms and electrons and gives the distances between atoms.
5. Using mathematical models, scientists draw a detailed map of atomic positions, bond angles and electron density, and so determine the overall structure of the molecule or crystal (e.g. naphthalene).

Q.28 [Short] Give the applications of X-ray crystallography.

- Determination of the double helix structure of DNA (its best-known use).
- Analysis of complex biomolecules such as enzymes and proteins.
- Pharmaceutical drug development: shows how a drug molecule attaches to its target.
- Analysis of minerals, synthetic crystals and nanomaterials, to improve materials for electronics, optics and construction.

31.4 FOSSIL FUELS

Q.29 [Long] What are fossil fuels? Describe their nature and uses.

Petroleum (crude oil), coal and natural gas are collectively called fossil fuels. **Crude oil** is taken from underground reservoirs and refined by fractional distillation; its major products are naphtha, gasoline, kerosene and diesel. **Natural gas** is mainly methane (CH_4) mixed with ethane, propane and butane; methane is used to produce ammonia (fertilizers) and methanol (solvents and fuel). **Coal** is still a petrochemical raw material where reserves are high; it gives benzene, toluene and xylene for dyes, resins and explosives.

Fossil fuel	Main uses
Petroleum	Transportation fuels (petrol, diesel), lubricants
Natural gas	Cooking, heating, electricity and fertilizer production
Coal	Electricity generation, cement and steel industries

Q.30 [Short] What is the impact of fossil fuels on the environment?

Fossil fuels are highly efficient sources of energy, but their combustion produces greenhouse gases such as carbon dioxide (CO_2) and others. This leads to air pollution, global warming and climate change. They are also non-renewable resources.

31.5 FRACTIONAL DISTILLATION OF PETROLEUM

Q.31 [Long] Explain the fractional distillation and refining of petroleum.

Crude oil is a complicated mixture of hydrocarbons. It is separated physically by **fractional distillation**, which depends on the difference in boiling points of the hydrocarbons.

1. The crude oil is heated in a furnace (above 370 °C) until it is vaporised.
2. The vapour passes into a fractionating column, a tall tower in which the temperature drops slowly going up.
3. As the vapour rises, the hydrocarbons condense at different heights according to their boiling points.
4. Heavier hydrocarbons with higher boiling points condense near the bottom; lighter hydrocarbons with low boiling points rise further.

Fraction (top to bottom)	Boiling range	Uses
Gases (e.g. LPG)	< 25 °C	Cooking and heating
Gasoline (petrol)	40–200 °C	Fuel for cars and bikes
Kerosene	150–275 °C	Jet fuel, lamps
Diesel	250–350 °C	Trucks, buses, generators
Lubricating oils	300–370 °C	Engine oil, grease
Residue (bitumen)	> 370 °C	Road surfacing, roofing

Quick Check 31.3

Q.32 [Short] a) If fossil fuels are non-renewable resources, why are they still heavily used worldwide?

Because they are highly efficient sources of energy and their fractions provide transport fuels, electricity, heating and the petrochemicals needed by industry. The existing vehicles, power plants and industries are built to run on them.

Q.33 [Short] b) Can XRD be used to find the structure of liquid?

No. X-ray crystallography needs a crystalline solid, in which atoms are arranged in regular layers that diffract X-rays at specific angles to give a clear pattern. In a liquid the particles are not in a fixed, regular arrangement, so no sharp diffraction pattern is formed.

31.6 PETROCHEMICALS

Q.34 [Short] Differentiate between petrochemicals and chemicals derived from them.

Petrochemicals are chemicals obtained primarily from crude oil and natural gas by refining and transformation, e.g. ethene, propene, benzene, toluene, xylene and butadiene. **Derivatives of petrochemicals** are the synthetic chemicals or substances manufactured from petrochemicals by chemical transformations such as polymerization, oxidation or substitution, e.g. poly(ethene), polystyrene, TNT and synthetic rubber. They are used in packaging, construction, textiles, electronics and medicine.

Q.35 [Short] Identify the raw materials for the petrochemical industry.

The raw materials of the petrochemical industry are the fossil fuels: crude oil (giving naphtha and other fractions), natural gas (methane with ethane, propane and butane) and coal (giving benzene, toluene and xylene).

Q.36 [Long] Identify the major petrochemicals, their derivatives and applications.

Petrochemical	Derived chemical / product	Uses
Ethene	Poly(ethene) (plastic)	Bags, bottles, packaging
Propene	Poly(propene)	Containers, automotive parts
Benzene	Styrene → polystyrene / poly(phenylethene)	Disposable cups, insulation
Toluene	TNT (trinitrotoluene)	Explosives
Xylene	Polyester fibres	Textiles, clothing (fleece, jerseys), upholstery
Butadiene	Synthetic rubber (e.g. styrene butadiene rubber)	Tires, footwear

Q.37 [Long] What is cracking? Compare thermal and catalytic cracking.

Cracking is the thermal dissociation of large hydrocarbon molecules (typically naphtha or heavy oils) into smaller, more useful molecules such as alkenes. It is necessary because ethene and propene are the major raw materials for plastics and artificial rubber.

Thermal cracking	Catalytic cracking
No catalyst is used.	A catalyst is used: zeolites (aluminosilicates) or silica-alumina.
Very high temperature (450–750 °C) and high pressure.	Lower temperature (about 500 °C) and lower pressure.

Thermal cracking	Catalytic cracking
Needs a high energy input.	The catalyst lowers the activation energy; energy is saved.
Gives a high percentage of alkenes (important for plastics).	More efficient, with better control over the final products; more common in modern refineries.

$$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4 \text{ (thermal cracking of decane)}$$
$$\text{Naphtha (heavy hydrocarbons)} \rightarrow C_2H_4 + C_3H_6 \text{ (catalyst)}$$

Q.38 [Short] How does a catalyst increase the rate of a reaction while remaining unchanged?

A catalyst provides an alternative reaction path with a lower activation energy, so more reacting molecules have enough energy to react and the rate increases. The catalyst takes part in the reaction steps but is regenerated, so it remains chemically unchanged at the end. Example: in catalytic cracking, zeolites lower the activation energy so that cracking occurs at a lower temperature (about 500 °C) and pressure, saving energy.

Q.39 [Long] What is reforming? Explain the reactions that take place during reforming.

Reforming is a refining process that rearranges the molecular structure of hydrocarbons: straight-chain alkanes are rearranged into branched or cyclic (aromatic) structures, without breaking them into smaller molecules. Its purpose is to improve fuel quality by converting low-octane straight chains into high-octane branched or aromatic fuels, which avoids engine knocking.

1. Isomerisation: straight-chain hydrocarbons are converted into branched-chain isomers of higher octane number.

$$C_6H_{14} \rightarrow \text{iso-}C_6H_{14} \text{ (n-hexane to iso-hexane)}$$

2. Cyclization: straight-chain hydrocarbons are converted into cyclic compounds.

$$C_6H_{14} \rightarrow C_6H_{12} + H_2 \text{ (hexane to cyclohexane)}$$

3. Dehydrocoupling (formation of aromatic hydrocarbons): hydrogen is lost to give aromatic compounds such as BTX (benzene, toluene, xylene), the building blocks of plastics, dyes and synthetic fibres. Hydrogen gas is a valuable by-product.

$$C_6H_{14} \rightarrow C_6H_6 + 4H_2 \text{ (Pt catalyst/heat)}$$

The textbook labels the last equation "(cyclohexane to benzene)", but as written ($C_6H_{14} \rightarrow C_6H_6 + 4H_2$) it is hexane to benzene; cyclohexane to benzene would be $C_6H_{12} \rightarrow C_6H_6 + 3H_2$.

Quick Check 31.4

Q.40 [Short] **a) Why is catalytic cracking more commonly used in modern petroleum refineries?**

The catalyst (zeolites or silica-alumina) lowers the activation energy, so cracking takes place at a lower temperature (about 500 °C) and pressure. This saves energy and gives better control over the final products; it is more efficient.

Q.41 [Short] **b) If more alkenes such as ethene and propene are desired to be produced from higher hydrocarbons, such as diesel or kerosine, which cracking method should be preferred and why?**

Thermal cracking, because it normally generates a high percentage of alkenes.

Q.42 [Short] **c) During reforming, hexane is converted into benzene and hydrogen. Explain why hydrogen can be considered a valuable by-product.**

Hydrogen is itself a useful substance: it is a clean fuel with a very high specific energy (120 MJ/kg) and it is needed in industry, e.g. for making ammonia for fertilizers. So it can be collected and used or sold instead of being wasted.

31.7 ENERGY DENSITY AND SPECIFIC ENERGY

Q.43 [Short] **Differentiate between energy density and specific energy.**

Energy density	Specific energy
Energy stored per unit volume of a fuel.	Energy contained per unit mass of a fuel.
Unit: megajoules per litre (MJ/L).	Unit: megajoules per kilogram (MJ/kg).
Important for stationary applications and storage (power plants, generators).	Important where weight is critical (aviation, spacecraft).
Example: diesel (~35.8 MJ/L) is more energy dense than gasoline (~34.2 MJ/L).	Example: hydrogen has a very high specific energy (120 MJ/kg), although its energy density is low.

Q.44 [Short] **Why is CNG suitable for local commuting but not for long journeys?**

CNG used in transport in Pakistan is very economical, but it has a lower energy density, so vehicles must carry more storage volume than for petrol. Therefore CNG is suitable for local commuting but not ideal for long journeys.

31.8 NUCLEAR ENERGY AND SOLAR ENERGY

Q.45 [Short] **What is nuclear energy?**

Nuclear energy is the energy released by the nuclei of atoms, either when heavy nuclei are split (fission) or when light nuclei are combined (fusion). These reactions liberate extremely large quantities of energy compared with chemical reactions.

Q.46 [Long] Describe nuclear fission, with its advantages and drawbacks.

Fission is the process in which a heavy nucleus such as uranium-235 or plutonium-239 is split into two lighter nuclei with the production of energy, neutrons and radiation. The neutrons produced can cause further fission, giving a **chain reaction**.

$$^{235}U + {}^{1}n \rightarrow {}^{141}Ba + {}^{92}Kr + 3\,{}^{1}n + \text{energy}$$

Advantages: nuclear power plants use fission extensively to produce electricity; a very large amount of energy is obtained from a small quantity of fuel. **Drawbacks:** it produces radioactive waste that must be handled with care for a long time, and there is the danger of nuclear accidents such as Chernobyl (1986) and Fukushima (2011), which seriously harmed the environment and human health. At Fukushima, an earthquake and tsunami caused reactor accidents and more than 150,000 citizens were evacuated.

Q.47 [Short] What is nuclear fusion? What is its energy significance?

Fusion is the process in which light nuclei combine to form a heavier nucleus with the release of a very large amount of energy. Example: nuclei of hydrogen isotopes (deuterium and tritium) fuse to form helium and a neutron. Fusion is the source of the Sun's energy; it gives more energy per unit mass than fission and little long-lived radioactive waste, but it needs extremely high temperatures, so it is not yet used in power plants.

$$^{2}H + {}^{3}H \rightarrow {}^{4}He + {}^{1}n + \text{energy}$$

The chapter heading names fusion but the text only defines it ("light nuclei are combined"); the example and details here are from general knowledge.

Q.48 [Long] Explain the working and significance of solar panels.

A solar PV (photovoltaic) system is a clean energy system in which solar panels, typically made of silicon semiconducting material, convert sunlight into electricity by the **photovoltaic effect**.

1. Sunlight falling on the panel activates electrons in the material, producing direct current (DC).
2. An inverter converts the DC into alternating current (AC), used in homes, schools, offices and industries.
3. Batteries store electricity for use at night or on cloudy days; they also help to stabilise voltage and supply surge current.
4. Electrical accessories, wires, appliances and mounting structures make the system effective and safe.

Solar energy is environmentally friendly and renewable, so it helps the fight against climate change and pollution. Solar panels are cost-effective in the long run because they need little maintenance once installed. Energy shortages, high fuel prices and climate problems show the need to switch to renewable sources like solar, hydro and wind energy.

Q.49 [Short] Explain the environmental impact of energy consumption and the need to reduce the carbon footprint.

Burning fossil fuels for energy releases carbon dioxide and other greenhouse gases, which cause air pollution, global warming and climate change. Reducing our **carbon footprint** (the amount of carbon dioxide released by our activities) is therefore important. This is done by saving energy and moving towards sustainable, renewable sources such as solar, hydro and wind energy.

Q.50 [Long] Evaluate fossil, nuclear and solar energy on cost, efficiency, impact and sustainability.

Source	Cost	Efficiency / output	Impact	Sustainability
Fossil fuels	Fuel must be bought continuously; prices are high	Highly efficient energy sources	Greenhouse gases, air pollution, global warming	Non-renewable
Nuclear fission	Expensive plants and waste handling	Very high energy from a small quantity of fuel	Radioactive waste; risk of accidents	Limited fuel, but long-lasting
Solar	Cost-effective in the long run; little maintenance	Depends on sunlight; needs batteries	Clean; no pollution in use	Renewable

The textbook does not give this comparison as a table; it is assembled from statements in sections 31.4, 31.8 and the solar energy section.

Quick Check 31.5

Q.51 [Short] a) Why are batteries included in many solar energy systems?

To store electricity for use at night or on cloudy days; they also help to stabilise the voltage and provide surge current.

Q.52 [Short] b) Besides power generation, where else is nuclear fission used?

In nuclear weapons (atomic bombs), in the reactors of nuclear submarines and ships, and in research reactors that produce radioisotopes for medicine and industry.

This answer is from general knowledge; the chapter mentions only electricity generation.

Q.53 [Short] c) Identify the missing component in the reaction: $^{235}U + n \rightarrow$ _____ + _____ + 3n + energy

$$^{235}U + {}^{1}n \rightarrow {}^{141}Ba + {}^{92}Kr + 3\,{}^{1}n + \text{energy}$$

The missing components are barium-141 and krypton-92.

Q.54 [Short] d) Why do nuclear reactions release much more energy than chemical reactions?

Chemical reactions involve only the electrons (breaking and making of bonds). Nuclear reactions involve changes in the nucleus, where the forces holding the particles are far stronger; a small amount of mass is converted into a very large amount of energy.

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